

Technology Stack

This document describes all technologies, libraries, and tools selected for the OptiCrop project, along with the rationale for each choice.

Programming Language

Language	Python 3.11
Rationale	Industry standard for ML/data science; rich ecosystem; cross-platform
Version	Python 3.11.x (minimum 3.10 required)

Machine Learning Stack

Library	Version	Purpose	Rationale
scikit-learn	1.4.x	Model training, evaluation, preprocessing	Gold standard ML library; RF, KNN, LR, Decision Tree
pandas	2.1.x	Data loading, EDA, feature engineering	DataFrame API ideal for tabular crop datasets
NumPy	1.26.x	Numerical computation, array operations	Fast array math for feature vectors
Matplotlib	3.8.x	Model evaluation plots (ROC, confusion matrix)	Compatible with Jupyter for visual EDA
Seaborn	0.13.x	Heatmaps, correlation plots for EDA	Declarative API for statistical plots
Pickle	stdlib	Model & scaler serialisation	Native Python; zero additional dependency

WebFramework & Frontend

Technology	Version	Role
Flask	3.0.x	Python micro web framework routing, request handling, Jinja2 templating
Jinja2	3.1.x	HTML templating engine dynamic result rendering within Flask
HTML5	5	Semantic markup for input forms and result pages
CSS3	3	Custom styling for form layout, colours, and typography
Bootstrap	5.3.x	Responsive grid, form components, and mobile-first design
JavaScript	ES2022	Client-side input validation and dynamic UI interactions

Development & DevOps

IDE Jupyter Notebook (EDA & model training) + VS Code (Flask app development)

Version Control: Git 2.x + GitHub repository hosting, branch strategy, pull requests

Environment Python virtualenv (venv) isolates project dependencies

Testing Pytest unit tests for preprocessing functions and prediction engine

Linting: flake8 PEP8 compliance; Black auto-formatter

Deployment Flask dev server (dev) | Gunicorn WSGI (production)